# OLIST Delivery Performance & Customer Dissatisfaction Analysis

Final Stakeholder Presentation

**Objective**:
Assess whether delivery delays are associated with customer dissatisfaction, identify where the issue is most relevant, and define the operational data needed for root-cause investigation.

**Scope**:
Delivered orders linked to valid customer reviews.
Delay analysis based on actual delivery date versus estimated delivery date.
São Paulo follow-up based on seller-origin ZIP/city areas and approximate ZIP-prefix coordinates.
Negative reviews defined as review score ≤ 2.

**Method**:
Order-level analysis using PostgreSQL outputs imported into Power BI.
The analysis identifies associations and follow-up priorities, not confirmed root causes.

**Dataset**: OLIST Brazilian E-Commerce
**Tools**: PostgreSQL, DBeaver, Power BI
**Status**: Final presentation — ready for stakeholder review

# Executive / Operational Summary

Delivery delays are a minority of reviewed delivered orders, but they are strongly associated with customer dissatisfaction and require targeted operational follow-up.

## Key findings

1. Late deliveries represent a small share of reviewed delivered orders, but show a much higher negative review rate.

2. São Paulo is the main delivered-order volume driver and should be analyzed separately before defining operational actions.

3. Candidate São Paulo ZIP/city areas help prioritize geographic follow-up, but do not confirm root cause.

4. Approval-to-carrier timing suggests a possible pre-carrier operational signal for late deliveries

## Operational meaning

The issue should not be treated as a general delivery-volume problem.

The priority is to isolate where late deliveries occur, whether they are linked to seller-origin areas, and whether delays start before carrier handoff.

The current dataset supports prioritization, not root-cause confirmation.

## Decision / next step

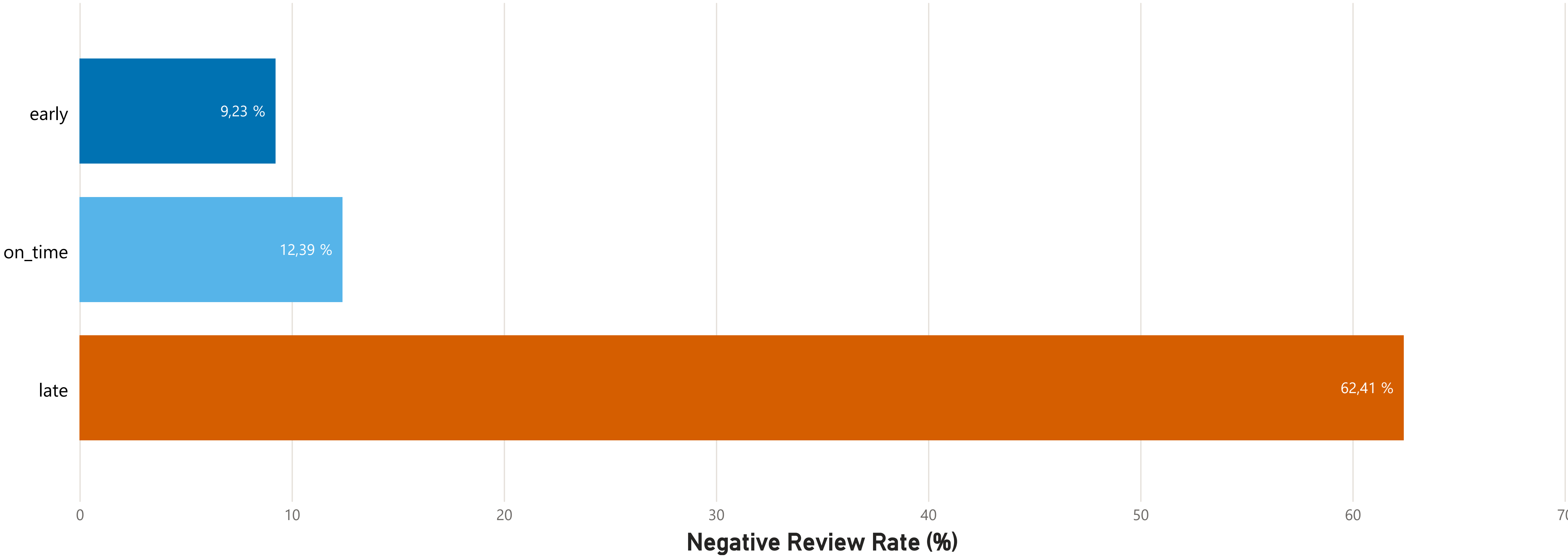
Use São Paulo as the priority follow-up area.

Validate the delay signal with operational data on seller/store mapping, carrier handoff, pickup/dispatch timing, backlog, capacity, stock availability, and staffing.

Do not treat the current findings as confirmed root cause.

# Small Delay Segment, High Dissatisfaction Impact

***Late deliveries account for only 6.65% of reviewed delivered orders, yet 62.41% of them receive negative reviews.***



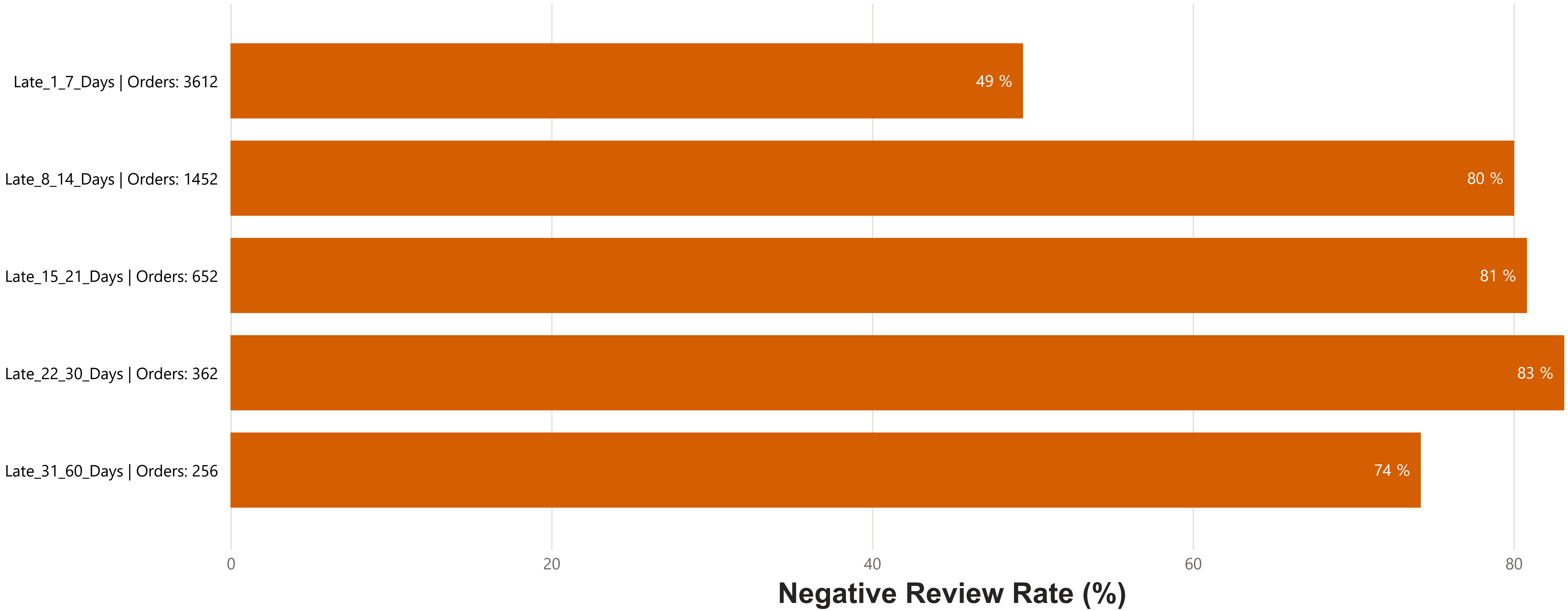

**Insight**:

Late deliveries represent only **6.65%** of reviewed delivered orders, but they show a negative review rate of **62.41%**.

This confirms that delivery delay is not a widespread volume issue, but when it occurs, it has a strong impact on customer dissatisfaction.

# Negative Review Rate by Delay Severity

The 1–7 day delay range has the highest volume, while longer delays show stronger dissatisfaction severity.

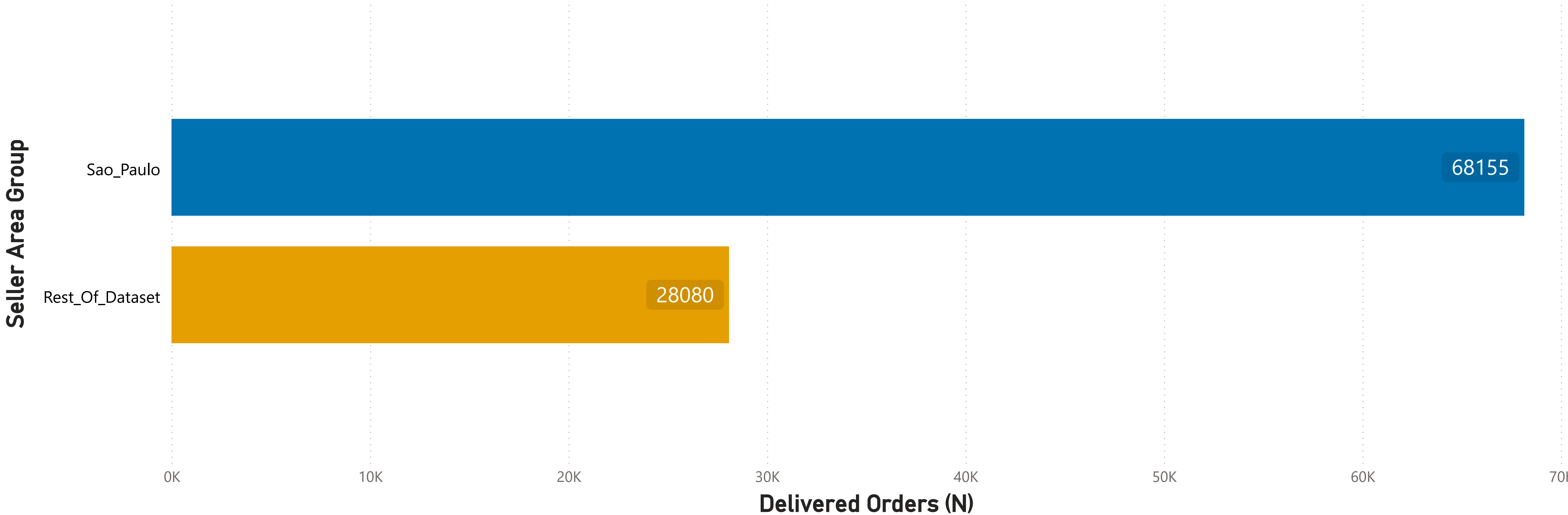



**Insight**:

The **1–7** day delay bucket has the largest volume with **3,612** orders and a **49.42%** negative review rate. Longer delays show higher severity, with negative review rates around **74%–83%**, but lower order volume.

# São Paulo Has Much Higher Delivered Order Volume

São Paulo represents the largest seller-origin volume, making it the priority area for follow-up analysis.



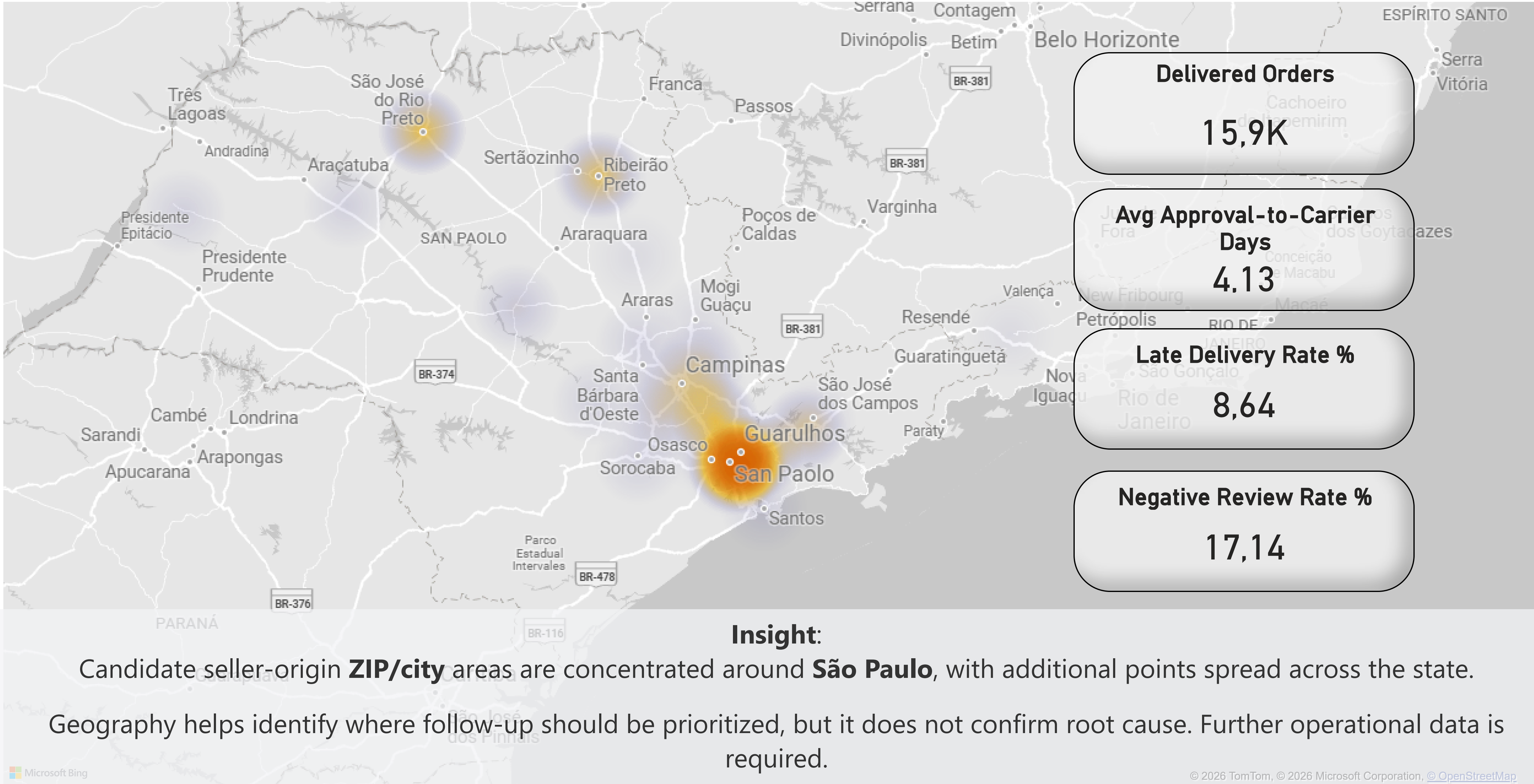

**Insight:**

São Paulo accounts for **68,155** delivered reviewed orders, around **143%** more than the rest of the dataset with **28,080** orders.

This confirms that São Paulo is the main volume driver and should be analyzed separately before defining operational actions.

# Geographic Concentration of São Paulo Candidate Seller-Origin Areas

Approximate ZIP-prefix locations used for follow-up analysis




**Delivered Orders**

15,9K

**Avg Approval-to-Carrier Days**

4.13

**Late Delivery Rate %**

8,64

**Negative Review Rate %**

17,14

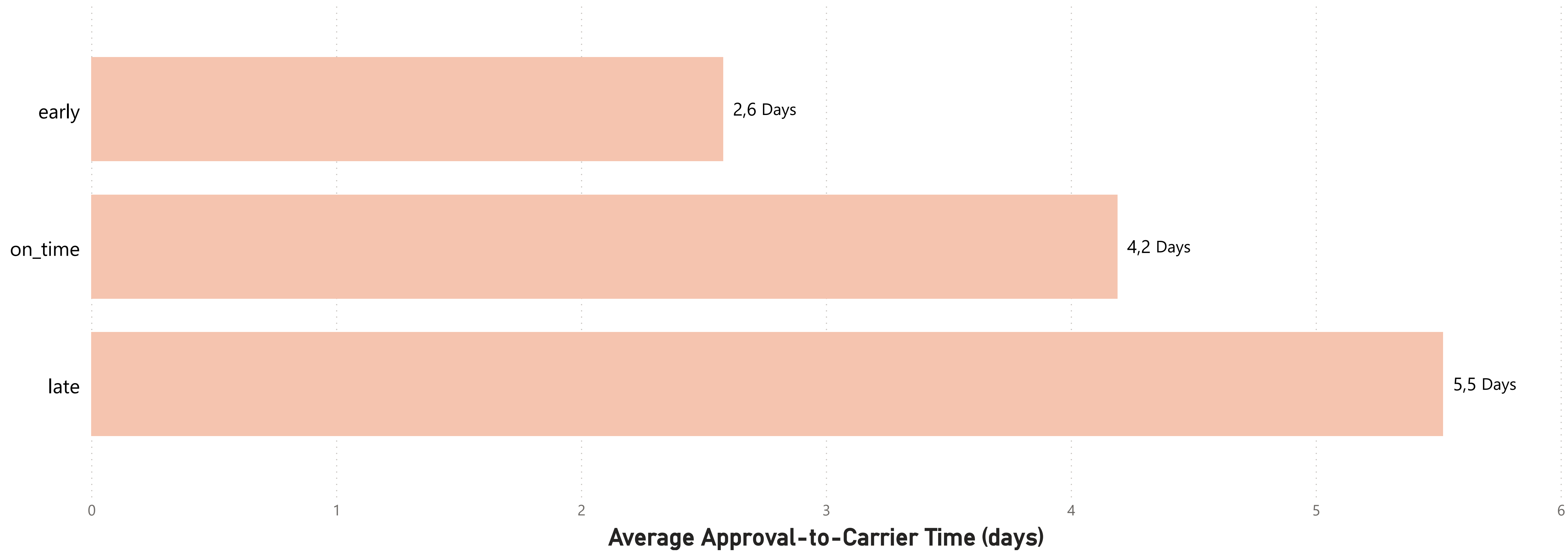
**Insight**:

Candidate seller-origin **ZIP/city** areas are concentrated around **São Paulo**, with additional points spread across the state.

Geography helps identify where follow-up should be prioritized, but it does not confirm root cause. Further operational data is required.

## Average Approval-to-Carrier Time by Delivery Status

Late deliveries show longer approval-to-carrier timing, suggesting a possible pre-carrier operational signal.




**Insight**:

Late deliveries show the **longest** approval-to-carrier **handoff time**, with an **average** of **5.52** days.

This suggests that part of the delay may start before the final carrier delivery phase, during order preparation, seller processing, warehouse handling, pickup scheduling, or carrier handoff.

# Data Needed / Limitations / Next Steps

## Limitations

The analysis identifies associations between delivery delays, seller-origin areas, approval-to-carrier timing, and customer dissatisfaction. It does not confirm root cause.

The geographic follow-up uses approximate ZIP-prefix coordinates. These coordinates do not represent exact seller, store, warehouse, or hub locations.

## Operational data needed

To validate the delay signal and investigate root cause, additional operational data is required:

- seller/store mapping
- operational clusters or fulfillment areas
- carrier assigned to each order
- pickup, dispatch, and handoff timestamps
- order preparation time
- stock availability
- backlog and workload
- capacity and staffing

## Next steps

1. Validate whether delays start before carrier handoff.
2. Compare São Paulo seller-origin areas with operational clusters and carrier routes.
3. Prioritize high-volume areas with higher late delivery and negative review rates.
4. Use operational data before defining corrective actions.